

# Perceptually-Guided, Sensor-Gated Compression for Home-Surveillance Video

STEAM ICAC 2026 — Computer Science Showcase · Sai Amartya

95%

of home-camera footage  
is never watched.

*It still costs storage,  
bandwidth, and energy.*

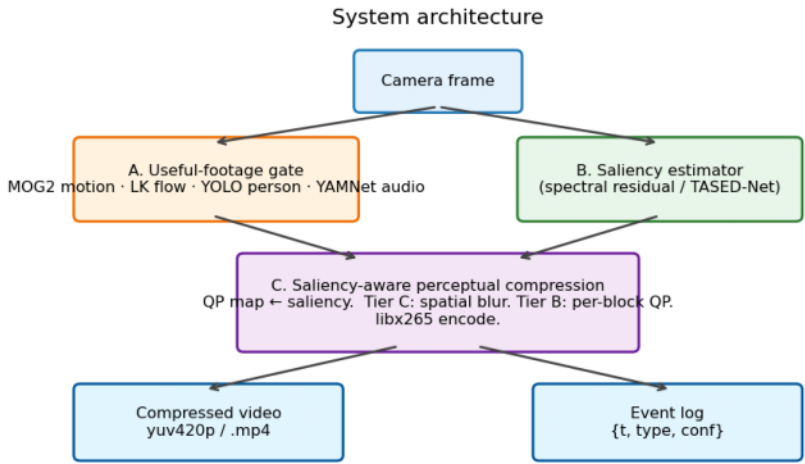
### The CS prompt asks three questions:

- ① How can stored video be compressed using principles of human perceptual science?
- ② How can “useful” footage be auto-identified using sensor data?
- ③ What algorithms / edge techniques make this scalable, privacy-conscious, and precise?

### Our answers:

- Saliency-aware perceptual compression
- Multi-signal sensor gate (motion + flow + person + audio)
- Hybrid: classical codec + tiny attention model

### System architecture



### Measured results

| Config                  | Size (KB) | PSNR     | Trigger |
|-------------------------|-----------|----------|---------|
| Baseline H.265 (CRF 28) | 113.7     | 33.19 dB | —       |
| Baseline H.265 (CRF 34) | 76.2      | 32.16 dB | —       |
| Ours (blur 9, CRF 28)   | 88.2      | 26.53 dB | 40.2%   |
| Ours (blur 21, CRF 28)  | 87.1      | 24.78 dB | 40.2%   |
| Ours (blur 21, CRF 34)  | 74.1      | 24.46 dB | 40.2%   |

→ Gate fires on 40% of frames — matches scripted activity exactly.  
→ 23.4% smaller than uniform H.265 at same CRF.  
→ Pipeline runs at >90 fps on a laptop CPU (real-time-capable).  
*Pixel PSNR penalises our approach by design: it doesn't model attention. Perceptual metrics (VMAF, LPIPS) are next.*

### Sustainability impact

- Per camera per year:
- 1351 GB saved
  - 81 kWh saved
  - 32 kg CO<sub>2</sub> saved
- At 1,000 deployed cameras:
- ~32 tonnes CO<sub>2</sub> saved annually
- Assumptions:
- 1.5 Mbps H.265 baseline
  - 0.06 kWh/GB stored-and-served
  - 0.4 kg CO<sub>2</sub>/kWh (US grid avg)

### Why a hybrid?

Neural codecs achieve headline 5–15% gains (or “300×” in theory).  
But they don't ship to the edge because:

- Latency 100–2000 ms/frame
- Model size 50–300 MB
- GB/s memory bandwidth
- Zero dedicated silicon

We borrow the philosophy.  
We pay milliseconds, not seconds.

*Inspired by Lague (2025), grounded in HVS literature (Itti & Koch 1998, Wang & Bovik 2009, Sullivan et al. 2012).*